

SQL JOIN Notes

Types: INNER, LEFT, RIGHT, FULL OUTER, CROSS, SELF.

INNER JOIN: Sirf matching records.

```
SELECT * FROM Employees e INNER JOIN Departments d ON  
e.DepartmentId=d.DepartmentId;
```

LEFT JOIN: Left table ke saare records, right me match na ho to NULL.

RIGHT JOIN: Right table ke saare records, left me match na ho to NULL.

FULL OUTER JOIN: Dono tables ke saare records.

CROSS JOIN: Har row $\times$ har row (Cartesian Product).

SELF JOIN: Same table ko khud se join.

Rapid Fire

1. JOIN: Multiple tables ka data combine karta hai.
2. Common column ke basis par hota hai.
3. INNER: Matching rows.
4. LEFT: Left sab rows.
5. RIGHT: Right sab rows.
6. FULL: Dono sab rows.
7. CROSS: Cartesian Product.
8. SELF: Same table join.
9. JOIN vs UNION: JOIN columns, UNION rows combine karta hai.
10. ON: matching condition ke liye.
11. CROSS JOIN me ON nahi lagta.
12. JOIN = INNER JOIN (default).
13. LEFT me NULL jab right match na ho.
14. RIGHT me NULL jab left match na ho.
15. Formula: Rows1 $\times$ Rows2.
16. Alias readability ke liye.
17. Performance: Index, ON clause, table size, optimization.
18. PK-FK par JOIN hota hai.
19. 2 se zyada tables bhi JOIN ho sakti hain.